

# Model for “graphene”

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## General Condition

- Basis type: 1g
- SAMB selection:
  - Type: [Q, G]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [ $A_{1g}$ ,  $A_{1u}$ ,  $A_{2g}$ ,  $A_{2u}$ ,  $B_{1g}$ ,  $B_{1u}$ ,  $B_{2g}$ ,  $B_{2u}$ ,  $E_{1g}$ ,  $E_{1u}$ ,  $E_{2g}$ ,  $E_{2u}$ ]
  - Spin (s): [0, 1]
- Atomic selection:
  - Type: [Q, G, M, T]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [ $A_{1g}$ ,  $A_{1u}$ ,  $A_{2g}$ ,  $A_{2u}$ ,  $B_{1g}$ ,  $B_{1u}$ ,  $B_{2g}$ ,  $B_{2u}$ ,  $E_{1g}$ ,  $E_{1u}$ ,  $E_{2g}$ ,  $E_{2u}$ ]
  - Spin (s): [0, 1]
- Site-cluster selection:
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [ $A_{1g}$ ,  $A_{1u}$ ,  $A_{2g}$ ,  $A_{2u}$ ,  $B_{1g}$ ,  $B_{1u}$ ,  $B_{2g}$ ,  $B_{2u}$ ,  $E_{1g}$ ,  $E_{1u}$ ,  $E_{2g}$ ,  $E_{2u}$ ]
- Bond-cluster selection:
  - Type: [Q, G, M, T]
  - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
  - Irrep.: [ $A_{1g}$ ,  $A_{1u}$ ,  $A_{2g}$ ,  $A_{2u}$ ,  $B_{1g}$ ,  $B_{1u}$ ,  $B_{2g}$ ,  $B_{2u}$ ,  $E_{1g}$ ,  $E_{1u}$ ,  $E_{2g}$ ,  $E_{2u}$ ]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: **false**

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## Group and Unit Cell

- Group: SG No. 191  $D_{6h}^1$   $P6/mmm$  [ hexagonal ]
- Associated point group: PG No. 191  $D_{6h}$   $6/mmm$  [ hexagonal ]
- Unit cell:
  - $a = 1.00000$ ,  $b = 1.00000$ ,  $c = 4.00000$ ,  $\alpha = 90.0$ ,  $\beta = 90.0$ ,  $\gamma = 120.0$
- Lattice vectors (conventional cell):
  - $\mathbf{a}_1 = [ 1.00000, 0.00000, 0.00000 ]$
  - $\mathbf{a}_2 = [ -0.50000, 0.86603, 0.00000 ]$
  - $\mathbf{a}_3 = [ 0.00000, 0.00000, 4.00000 ]$

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**Symmetry Operation**


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Table 1: Symmetry operation

| #  | SO                | #  | SO                 | #  | SO                 | #  | SO                 | #  | SO                 |
|----|-------------------|----|--------------------|----|--------------------|----|--------------------|----|--------------------|
| 1  | $\{1 0\}$         | 2  | $\{3_{001}^+ 0\}$  | 3  | $\{3_{001}^- 0\}$  | 4  | $\{2_{001} 0\}$    | 5  | $\{6_{001}^- 0\}$  |
| 6  | $\{6_{001}^+ 0\}$ | 7  | $\{2_{110} 0\}$    | 8  | $\{2_{100} 0\}$    | 9  | $\{2_{010} 0\}$    | 10 | $\{2_{1-10} 0\}$   |
| 11 | $\{2_{120} 0\}$   | 12 | $\{2_{210} 0\}$    | 13 | $\{-1 0\}$         | 14 | $\{-3_{001}^+ 0\}$ | 15 | $\{-3_{001}^- 0\}$ |
| 16 | $\{m_{001} 0\}$   | 17 | $\{-6_{001}^- 0\}$ | 18 | $\{-6_{001}^+ 0\}$ | 19 | $\{m_{110} 0\}$    | 20 | $\{m_{100} 0\}$    |
| 21 | $\{m_{010} 0\}$   | 22 | $\{m_{1-10} 0\}$   | 23 | $\{m_{120} 0\}$    | 24 | $\{m_{210} 0\}$    |    |                    |

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**Harmonics**


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Table 2: Harmonics

| # | symbol                     | irrep.   | rank | X      | multiplicity | component | symmetry                         |
|---|----------------------------|----------|------|--------|--------------|-----------|----------------------------------|
| 1 | $\mathbb{Q}_0(A_{1g})$     | $A_{1g}$ | 0    | $Q, T$ | -            | -         | 1                                |
| 2 | $\mathbb{G}_1(A_{2g})$     | $A_{2g}$ | 1    | $G, M$ | -            | -         | $z$                              |
| 3 | $\mathbb{Q}_3(B_{1u})$     | $B_{1u}$ | 3    | $Q, T$ | -            | -         | $\frac{\sqrt{10}y(3x^2-y^2)}{4}$ |
| 4 | $\mathbb{Q}_3(B_{2u})$     | $B_{2u}$ | 3    | $Q, T$ | -            | -         | $\frac{\sqrt{10}x(x^2-3y^2)}{4}$ |
| 5 | $\mathbb{Q}_{1,1}(E_{1u})$ | $E_{1u}$ | 1    | $Q, T$ | -            | 1         | $x$                              |
| 6 | $\mathbb{Q}_{1,2}(E_{1u})$ |          |      |        |              | 2         | $y$                              |

*continued ...*

Table 2

| # | symbol                     | irrep.   | rank | X      | multiplicity | component | symmetry                       |
|---|----------------------------|----------|------|--------|--------------|-----------|--------------------------------|
| 7 | $\mathbb{Q}_{2,1}(E_{2g})$ | $E_{2g}$ | 2    | $Q, T$ | -            | 1         | $\frac{\sqrt{3}(x-y)(x+y)}{2}$ |
| 8 | $\mathbb{Q}_{2,2}(E_{2g})$ |          |      |        |              | 2         | $-\sqrt{3}xy$                  |

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Basis in full matrix

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Table 3: dimension = 2

| # | orbital@atom(SL)    | # | orbital@atom(SL)    |
|---|---------------------|---|---------------------|
| 0 | $ p_z\rangle @C(1)$ | 1 | $ p_z\rangle @C(2)$ |

Table 4: Atomic basis (orbital part only)

| orbital       | definition |
|---------------|------------|
| $ p_x\rangle$ | $x$        |
| $ p_y\rangle$ | $y$        |
| $ p_z\rangle$ | $z$        |

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SAMB: 11 (all 11)

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- **C : 'C' site-cluster**

- \* bra:  $\langle p_z |$

- \* ket:  $|p_z\rangle$

- \* wyckoff: **2c**

$$\boxed{\text{z1}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(s)}(A_{1g})$$

$$\boxed{\text{z4}} \quad \mathbb{Q}_3^{(c)}(B_{1u}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(s)}(B_{1u})$$

- **C;C\_001\_1 : 'C'-'C' bond-cluster**

- \* bra:  $\langle p_z |$

- \* ket:  $|p_z\rangle$

- \* wyckoff: **3a03f**

$$\boxed{\text{z2}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z8}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

- **C;C\_002\_1 : 'C'-'C' bond-cluster**

- \* bra:  $\langle p_z |$

- \* ket:  $|p_z\rangle$

- \* wyckoff: **6b061**

$$\boxed{\text{z3}} \quad \mathbb{Q}_0^{(c)}(A_{1g}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_0^{(b)}(A_{1g})$$

$$\boxed{\text{z5}} \quad \mathbb{Q}_3^{(c)}(B_{1u}) = \mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_3^{(b)}(B_{1u})$$

$$\boxed{\text{z6}} \quad \mathbb{Q}_{1,1}^{(c)}(E_{1u}) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z7}} \quad \mathbb{Q}_{1,2}^{(c)}(E_{1u}) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{1,2}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{z10}} \quad \mathbb{Q}_{2,1}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,1}^{(b)}(E_{2g})}{2}$$

$$\boxed{\text{z11}} \quad \mathbb{Q}_{2,2}^{(c)}(E_{2g}) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_{1g})\mathbb{Q}_{2,2}^{(b)}(E_{2g})}{2}$$

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### Atomic SAMB

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- bra:  $\langle p_z |$
- ket:  $|p_z\rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_{1g}) = [1]$$

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### Cluster SAMB

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- Site cluster

\*\* Wyckoff: 2c

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_{1g}) = \left[ \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right]$$

$$\boxed{\text{y2}} \quad \mathbb{Q}_3^{(s)}(B_{1u}) = \left[ \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

- Bond cluster

\*\* Wyckoff: 3a@3f

$$\boxed{\text{y3}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[ \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y4}} \quad \mathbb{T}_3^{(b)}(B_{1u}) = \left[ \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3} \right]$$

$$\boxed{\text{y5}} \quad \mathbb{T}_{1,1}^{(b)}(E_{1u}) = \left[ 0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2} \right]$$

$$\boxed{\text{y6}} \quad \mathbb{T}_{1,2}^{(b)}(E_{1u}) = \left[ \frac{\sqrt{6}i}{3}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y7}} \quad \mathbb{Q}_{2,1}^{(b)}(E_{2g}) = \left[ \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y8}} \quad \mathbb{Q}_{2,2}^{(b)}(E_{2g}) = \left[ 0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \right]$$

\*\* Wyckoff: **6b@61**

$$\boxed{\text{y9}} \quad \mathbb{Q}_0^{(b)}(A_{1g}) = \left[ \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y10}} \quad \mathbb{M}_1^{(b)}(A_{2g}) = \left[ \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y11}} \quad \mathbb{Q}_3^{(b)}(B_{1u}) = \left[ \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6}, -\frac{\sqrt{6}}{6} \right]$$

$$\boxed{\text{y12}} \quad \mathbb{T}_3^{(b)}(B_{2u}) = \left[ \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, \frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6}, -\frac{\sqrt{6}i}{6} \right]$$

$$\boxed{\text{y13}} \quad \mathbb{Q}_{1,1}^{(b)}(E_{1u}) = \left[ 0, -\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2} \right]$$

$$\boxed{\text{y14}} \quad \mathbb{Q}_{1,2}^{(b)}(E_{1u}) = \left[ \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y15}} \quad \mathbb{T}_{1,1}^{(b)}(E_{1u}) = \left[ \frac{\sqrt{3}i}{3}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y16}} \quad \mathbb{T}_{1,2}^{(b)}(E_{1u}) = \left[ 0, \frac{i}{2}, -\frac{i}{2}, 0, -\frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y17}} \quad \mathbb{Q}_{2,1}^{(b)}(E_{2g}) = \left[ \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{6}, -\frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y18}} \quad \mathbb{Q}_{2,2}^{(b)}(E_{2g}) = \left[ 0, \frac{1}{2}, -\frac{1}{2}, 0, \frac{1}{2}, -\frac{1}{2} \right]$$

$$\boxed{\text{y19}} \quad \mathbb{T}_{2,1}^{(b)}(E_{2g}) = \left[ 0, \frac{i}{2}, -\frac{i}{2}, 0, \frac{i}{2}, -\frac{i}{2} \right]$$

$$\boxed{\text{y20}} \quad \mathbb{T}_{2,2}^{(b)}(E_{2g}) = \left[ -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, -\frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

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## Site and Bond

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Table 5: Orbital of each site

| # | site | orbital       |
|---|------|---------------|
| 1 | C    | $ p_z\rangle$ |

Table 6: Neighbor and bra-ket of each bond

| # | head | tail | neighbor | head (bra) | tail (ket) |
|---|------|------|----------|------------|------------|
| 1 | C    | C    | [1,2]    | [p]        | [p]        |

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## Site in Unit Cell

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Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'C' (#1) site cluster (2c), -6m2

| SL | position ( $\mathbf{s}$ )    | mapping                            |
|----|------------------------------|------------------------------------|
| 1  | [ 0.33333, 0.66667, 0.00000] | [1,2,3,10,11,12,16,17,18,19,20,21] |

*continued ...*

Table 7

| SL | position ( $\mathbf{s}$ )    | mapping                         |
|----|------------------------------|---------------------------------|
| 2  | [ 0.66667, 0.33333, 0.00000] | [4,5,6,7,8,9,13,14,15,22,23,24] |

### Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'C'-'C' [1] (#1) bond cluster (3a@3f), ND,  $|\mathbf{v}|=0.57735$  (cartesian)

| SL | vector ( $\mathbf{v}$ )      | center ( $\mathbf{c}$ )      | mapping                    | head  | tail  | $\mathbf{R}$ (primitive) |
|----|------------------------------|------------------------------|----------------------------|-------|-------|--------------------------|
| 1  | [ 0.33333, 0.66667, 0.00000] | [ 0.50000, 0.00000, 0.00000] | [1,-4,-8,11,-13,16,20,-23] | (2,1) | (1,1) | [0,-1,0]                 |
| 2  | [-0.66667,-0.33333, 0.00000] | [ 0.00000, 0.50000, 0.00000] | [2,-5,-7,10,-14,17,19,-22] | (2,1) | (1,1) | [1,0,0]                  |
| 3  | [ 0.33333,-0.33333, 0.00000] | [ 0.50000, 0.50000, 0.00000] | [3,-6,-9,12,-15,18,21,-24] | (2,1) | (1,1) | [0,0,0]                  |

Table 9: 2-th 'C'-'C' [1] (#2) bond cluster (6b@6l), ND,  $|\mathbf{v}|=1.0$  (cartesian)

| SL | vector ( $\mathbf{v}$ )      | center ( $\mathbf{c}$ )      | mapping        | head  | tail  | $\mathbf{R}$ (primitive) |
|----|------------------------------|------------------------------|----------------|-------|-------|--------------------------|
| 1  | [ 1.00000, 0.00000, 0.00000] | [ 0.83333, 0.66667, 0.00000] | [1,-11,16,-20] | (1,1) | (1,1) | [-1,0,0]                 |
| 2  | [ 0.00000, 1.00000, 0.00000] | [ 0.33333, 0.16667, 0.00000] | [2,-10,17,-19] | (1,1) | (1,1) | [0,-1,0]                 |
| 3  | [-1.00000,-1.00000, 0.00000] | [ 0.83333, 0.16667, 0.00000] | [3,-12,18,-21] | (1,1) | (1,1) | [1,1,0]                  |
| 4  | [-1.00000, 0.00000, 0.00000] | [ 0.16667, 0.33333, 0.00000] | [4,-8,13,-23]  | (2,1) | (2,1) | [1,0,0]                  |

*continued ...*



Table 9

| SL | vector ( $\boldsymbol{v}$ )   | center ( $\boldsymbol{c}$ )  | mapping       | head  | tail  | $\boldsymbol{R}$ (primitive) |
|----|-------------------------------|------------------------------|---------------|-------|-------|------------------------------|
| 5  | [ 0.00000, -1.00000, 0.00000] | [ 0.66667, 0.83333, 0.00000] | [5,-7,14,-22] | (2,1) | (2,1) | [0,1,0]                      |
| 6  | [ 1.00000, 1.00000, 0.00000]  | [ 0.16667, 0.83333, 0.00000] | [6,-9,15,-24] | (2,1) | (2,1) | [-1,-1,0]                    |